UNIVERSITY OF INFORMATION TECHNOLOGY AND SCIENCES



SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

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SRS Topic : BloomVerse

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1. Introduction

1.1. Purpose

Urban gardening is growing fast, but plant lovers usually have to use many separate platforms. One for buying supplies, one for finding out why a plant is sick, one for giving away extra plants, and another for delivery. BloomVerse puts all of these into one web system.

BloomVerse has four main parts. A categorized shop (Plants, Flowers, Seeds, Fertilizers, Gardening Tools). A Plant Doctor that checks a leaf photo and suggests a fix. A Swap Hub where members give away or swap plants. And a Green Route planner that helps delivery riders finish their orders with a short, planned route.

1.2 Document Conventions

A. The entire document justified.

B. Convention for Main title

- Font face: Times New Roman
- Font style: Bold
- Font Size: 14

C. Convention for Sub title

- Font face: Times New Roman
- Font style: Bold
- Font Size: 14

D. Convention for body

- Font face: Times New Roman
- Font style: Normal
- Font Size: 12

1.3 Scope of Development Project

BloomVerse is a web application built with HTML, CSS, JavaScript, PHP and a MySQL database. With it, users can:

- Browse and buy products from five sections: Plants, Flowers, Seeds, Fertilizers and Gardening Tools, with search, category filter and an Eco-Certified filter.
- Upload a leaf photo to the Plant Doctor and get a health index, the likely problem and simple care steps.
- Give away or swap plants through the Swap Hub, with an optional photo.
- Place orders with Cash on Delivery and download a PDF receipt.
- Rate and review products they have received, and flag bad reviews.

Riders can go on shift, see new orders, accept them, and follow a planned route on a map. Admins manage products, riders, orders, Eco-Certified labels and flagged reviews.

Online payment, live GPS tracking of riders and a mobile app are outside the scope of this version.

1.4 Definitions, Acronyms and Abbreviations

Term	Meaning
Admin	System administrator who manages products, orders, riders, swap posts and reviews.
User / Customer	Registered person who shops, swaps plants and uses the Plant Doctor.
Delivery Rider	Person who delivers orders and uses the Green Route module.
Plant Doctor	Rule-based leaf checker. It reads the colors of the leaf photo (green, yellow, brown/dark) and decides the likely problem. It does not use machine learning.
Depot	Fixed starting point of every delivery route.
Shift	A rider is either on shift (can accept orders and see the route) or off shift.

Term	Meaning
COD	Cash on Delivery. The customer pays the rider when the order arrives.
DBMS	Database Management System (MySQL).
Leaflet	Open-source JavaScript library used to show the map.
OSRM	Open-Source Routing Machine, a free service that gives road distance and travel time.
2-opt	A method that improves a route by reversing parts of it when that makes the total distance shorter.
Eco-Certified	Label for organic/eco-friendly products, which only an admin can give.
Moderation Queue	Admin list of flagged reviews waiting for action.
SRS	Software Requirements Specification.
Figma	Design tool used for the UI/UX prototype.

2. Overall Description

2.1 Product Perspective

BloomVerse is a client-server web system.

- Frontend: HTML5, CSS3 and JavaScript. Leaflet shows the map and PDF creates the receipt.
- Backend: PHP handles login, orders, the Plant Doctor logic, swap posts and the route data. Database access uses PDO prepared statements.
- Database: MySQL stores users, products, reviews, orders, order items, swap posts, swap interests and health logs.
- External services: OpenStreetMap map tiles and the OSRM public server for road routes. If they are not reachable, the system still shows a straight-line route.

2.2 Product Functions

1. User registration and login: Sign up, login and logout. Passwords are stored with bcrypt (PHP password_hash). Access depends on role: Admin, User or Rider.
2. Categorized shop: Plants, Flowers, Seeds, Fertilizers and Tools with search, filter, ratings and reviews.
3. Plant Doctor: Leaf photo (JPG or PNG, up to 8 MB) is analyzed and the user gets a health index, the issue, care steps and scan history.
4. Swap Hub: Users list a plant as Free adoption or Swap, add a photo, and show interest in other listings. Owners and admins can remove a listing.
5. Cart and checkout: Cart is kept in the browser session. The order needs name, phone, full address and delivery area. Stock is checked when the order is placed, and a PDF receipt can be downloaded.
6. Green Route: Rider goes on shift, accepts new orders, and gets a planned route on a map with stop list, time, distance and daily statistics.
7. Admin panel: Statistics, recent orders, rider assignment, Eco-Certified labels, flagged reviews, add product and make a user a rider.
8. Database management: Data is stored in MySQL with foreign keys for relations.

2.3 User Classes and Characteristics

2.3.1 Administrator: Good technical knowledge. Manages products, riders, orders, reviews and swap posts.

2.3.2 User / Customer: Basic web knowledge. Buys products, scans leaves, lists or adopts plants and tracks orders.

2.3.3 Delivery Rider: Uses a phone or browser while working. Goes on shift, accepts orders, follows the route and marks orders delivered.

2.3.4 System perspective on users: PHP sessions and server-side role checks make sure each user only reaches the modules allowed for their role.

2.4 Operating Environment

BloomVerse runs in modern browsers on desktops, tablets and smartphones. The server runs Apache, PHP and MySQL (XAMPP during development).

2.5 Assumptions and Dependencies

Assumptions:

- Users have internet access and a camera or gallery to upload plant photos.
- Riders use an internet-connected smartphone.

Dependencies:

- Active MySQL connection and a web server with PHP and the GD image extension turned on.
- Internet access for the map (OpenStreetMap), road routing (OSRM) and the PDF library.

2.6 Requirements

Functional requirements:

- Signup, login, logout and role-based access.
- Browse, search and filter products by section.
- Place a COD order and download a receipt.
- Upload a leaf photo and get a diagnosis.
- Create, view, remove and show interest in swap listings.
- Rider shift, order accept, route planning and delivery status update.
- Admin management of products, orders, riders and reviews.

Non-functional requirements:

- Normal pages should load in under 3 seconds on a normal broadband connection.
- Plant Doctor result should come back in under 5 seconds.
- Passwords must be hashed with bcrypt, and all database queries must use prepared statements.
- The design must work on both phone and desktop screens.

Hardware and software requirements:

- Frontend: HTML, CSS, JavaScript (Leaflet, jsPDF)
- Backend: PHP 7.4 or higher with GD extension
- Database: MySQL
- Development tool: XAMPP

2.7 Data Requirements

Inputs: registration data, product searches, cart items, delivery details and area, leaf photos, swap listing details and rider actions (shift, accept, deliver).

Outputs: product pages, order receipt, health report (health index, issue, steps), swap listings, and the Green Route map with stop order, distance and time.

3. External Interface Requirements

User interface: Clean responsive pages. Customers see shop, cart, orders, Plant Doctor and Swap Hub. Riders also see Green Route. Admins also see the Admin dashboard.

Software interface: Browser with PHP and MySQL on the server. Leaflet.js with OpenStreetMap tiles shows the map. The OSRM public API gives road distance and travel time. PDF creates the receipt.

Communication interface: HTTP/HTTPS over TCP/IP. The frontend talks to the PHP API using JSON.

4. System Features

- Home page: Hero banner with Explore Now and Learn How It Works buttons, category cards and footer.
- Categorized shop: Product grid with category tabs, search, Eco-Certified filter and average rating.
- Reviews and ratings: Star rating and comment, only after the order is delivered. Any review can be flagged.
- Plant Doctor: Photo upload, health index, issue, care steps and last scans.
- Swap Hub: Listing cards with type (Free adoption or Swap), area, owner, interested count and a create-listing form.

- User authentication: Login, registration and logout.
- Cart and checkout: Cart, delivery details, Cash on Delivery and PDF receipt.
- My orders: Customer sees own orders and their status.
- Green Route Planner: See section 4.1.
- Admin dashboard: Statistic cards, recent orders with rider assignment, Eco-Certified buttons, moderation queue, add product and make rider.
- Responsive design: CSS grid and flexbox for phone and desktop.

4.1 Green Route Planner

Order status flow: Placed, then Out for delivery, then delivered.

Rider steps:

1. Rider logs in and turns the shift on. When off shift, the rider sees only the daily summary.
2. The page lists new orders waiting: orders with status Placed and no rider. Each shows area, address, amount and distance from the depot.
3. The rider presses accept for delivery. The order is given to that rider only (first rider to accept gets it) and its status becomes Out for delivery. An admin can also assign an order manually.
4. The system plans the route for the rider's accepted orders and shows it on the map with numbered stops.
5. For each stop the rider can call the customer, open navigation in Google Maps, and press *Mark delivered*. The route is planned again after every delivery.

Route planning algorithm:

1. Start from the depot.
2. *Nearest neighbor:* always go to the closest unvisited stop (straight-line distance by the haversine formula).
3. *2-opt improvement:* try reversing parts of the route and keep any change that makes the total distance shorter. Repeat until no better route is found.

4. *Road route:* ask OSRM for the real road line, distance and travel time. If OSRM is not available, the straight-line route is shown and time is estimated at 20 km/h.
5. Add 3 minutes handling time per stop to the estimated time.

Daily statistics shown to the rider: stops left, delivered today, cash collected today, route length, estimated time and CO2 saved.

CO2 saved is an estimate: (distance of the unplanned order minus distance of the planned route) x 100 g per km, assuming a petrol motorbike.

5. Other Non-functional Requirements

5.1 Performance Requirement

The system should work for many users at the same time without lag. Product listing and order queries should respond in under 1 second on the development server. Stock is checked inside a database transaction so two customers cannot buy the last item together.

5.2 Safety Requirement

Uploaded images must be checked before use. Only JPG and PNG files are accepted (file type checked on the server, not only the file name), size is limited, and swap photos are resized and saved again as JPEG. No uploaded file is ever executed on the server.

5.3 Security Requirement

- Passwords are hashed.
- All SQL uses PDO prepared statements to stop SQL injection.
- Text shown on pages is escaped to stop XSS.
- Role checks are done on the server for every protected action.
- If a logged-in account is deleted or its role changes, the session is updated on the next request.

5.4 Requirement Attributes

Every requirement has a unique ID, type, priority, source and implementation status.

ID	Requirement	Type	Priority	Status
FR-01	User signup, login and logout	Functional	High	Done
FR-02	Browse and search products by section	Functional	High	Done
FR-03	Cart and COD checkout with PDF receipt	Functional	High	Done
FR-04	Plant Doctor leaf analysis	Functional	High	Done
FR-05	Swap Hub listings and interest	Functional	Mediu m	Done
FR-06	Rider shift and accept order	Functional	High	Done
FR-07	Green Route planning on map	Functional	High	Done
FR-08	Mark order delivered	Functional	High	Done
FR-09	Admin dashboard and moderation	Functional	Mediu m	Done
NFR-01	Page load under 3 seconds	Non-functional	Mediu m	Planned
NFR-02	bcrypt passwords, prepared statements	Non-functional	High	Done
NFR-03	Safe image upload	Non-functional	High	Done

5.5 Business Rules

- Users must be registered and logged in to order products, post in the Swap Hub, or use the Plant Doctor.
- Free adoption listings have no price.
- Only customers who received a product (order status Delivered) can review it.
- Only an admin can mark a product Eco-Certified, and only an admin can turn a user into a rider.
- A rider must be on shift to accept orders.
- One order belongs to only one rider. If two riders press Accept together, only the first one gets it.

- A rider can see the route and customer phone only for orders that belong to that rider.
- Only the owner of a swap listing or an admin can remove it.

5.6 User Requirement

Users need an easy and good-looking website where they can buy plant supplies, check plant health, swap plants and get their orders delivered.

6. Other Requirements

6.1 Data and Category Requirements

The MySQL database has these tables:

Table	Main fields
users	id, name, email (unique), password (hash), role (admin / user / rider), on_shift
products	id, name, category (Plants / Seeds / Fertilizers / Flowers / Tools), price, stock, eco
reviews	id, product_id, user_id, rating, comment, flagged
orders	id, user_id, cust_name, phone, address, area, lat, lng, total, status (Placed / Out for delivery / Delivered), rider_id, created, delivered_at
order_items	order_id, product_id, qty, price
swap_posts	id, user_id, title, type (Free adoption / Swap), area, photo
swap_interests	post_id, user_id
health_logs	id, user_id, health_index, issue, created

Design notes: the cart is kept in the browser session, so it has no table. Roles and categories are stored as fixed values in the users and products tables. Flagged reviews are marked with the flagged column and shown in the moderation queue. Route data comes from the orders table (location, rider_id and status).

6.2 Glossary

- BloomVerse: Name of the integrated platform.
- Green Route: Module that plans a short delivery route for riders.

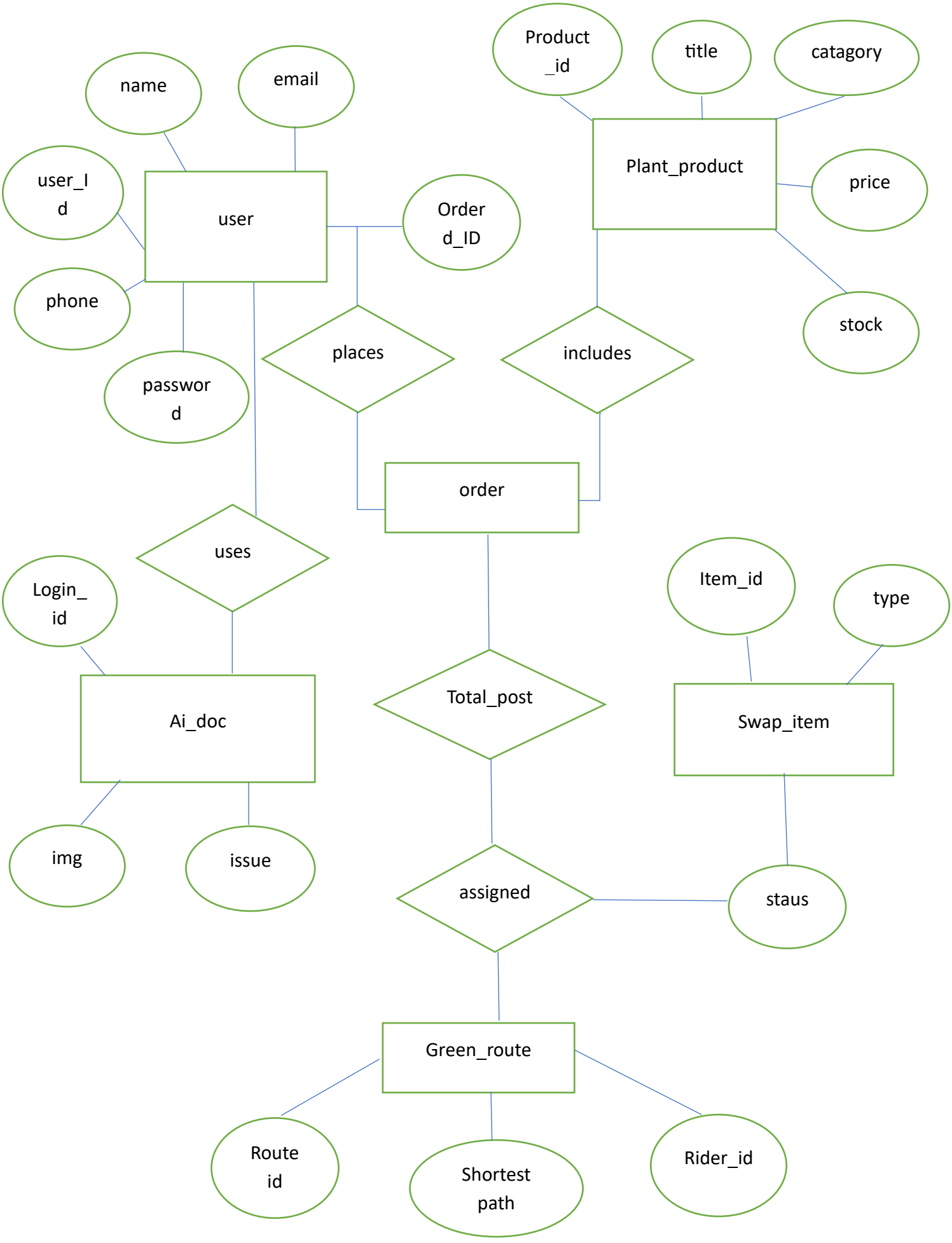
- Swap Hub: Community place for free plant adoption and swapping.
- Health Index: Percentage score showing how healthy a scanned leaf looks.
- Depot: Starting point of every route.

6.3 ER Diagram

An entity is a real-world object that we store data about. Each entity has attributes, a unique key and relationships with other entities. The main entities of BloomVerse are Users, Products, Orders, Order Items, Reviews, Swap Posts, Swap Interests and Health Logs. A rider is also a row in Users.

The relationships are:

Relationship	Cardinality
User places Order	1 : N
Rider (User) delivers Order	1 : N
Order includes Order Items	1 : N
Product appears in Order Items	1 : N
Order and Product (through Order Items)	M : N
User writes Review	1 : N
Product has Reviews	1 : N
User creates Swap Post	1 : N
User shows interest in Swap Post (through Swap Interests)	M : N
User has Health Logs (Plant Doctor scans)	1 : N


Product _id
title
catagory
name
email
Plant_product
user_I d
price
user
Order d_ID
phone
stock
places
includes
passwor d
order
uses
Login_ id
Item_id
type
Total_post
Ai_doc
Swap_item
img
issue
assigned
staus
Green_route
Route id
Shortest path
Rider_id

7. Conclusion

BloomVerse brings online shopping, plant health checking, plant swapping and planned delivery into one web system. It is built with HTML, CSS, JavaScript, PHP and MySQL, with Leaflet and OSRM for the Green Route map. The project shows how a simple web system can support eco-friendly habits, urban gardening and efficient delivery.